## UAV sources

**Local flight state**
Position / velocity / heading

**Private credits**
Private byte / credit balance
Balances remain local

**Protected attempts and retries**
Bounded retries / failure recovery

Reports

Controls

## Intermittent wireless service

**Uplink**
Data / registration / release

**Future service rights**
Owner / slot / capacity / expiry

**Downlink**
Grants / ACKs

**Paid bytes: data + control**

Received reports

Service events

Service rights

Grants / ACKs

## Ground monitoring platform

Paid grant/ACK events

**State monitor**
Received-message reassembly / estimation

History

H(t)

**Causal resource support**
Exact no-report marginals / causal inference

Supports

**Executable admission and ACK guard**
Resource allocation /
All remaining obligations /
Guarded ACK payments

Quality bounds

**Valid quality profile**
Validated model bounds /
Forward quality limits

Outstanding ACK debt (generated, not yet transmitted)

Received registration

**Complete registration**
reassembly + validation

Activate profile